**Controlled Gaussian-score simulations**

135 design cells | 1,000 repetitions per cell

**(a) Intercept shift**

$x \rightarrow s = x + 1$

Level at s; change relative to x

| | 0.5 | 1.0 | 2.0 |
|---|---|---|---|
| Decision loss (level at $s$, pp) | 7.7 | 10.2 | 12.4 |
| Headroom change | 0.0 | 0.0 | 0.0 |

Class separation, $d$

**(b) Positive scaling**

$s \rightarrow 0.2s$

Changes relative to shifted score s

| | 0.5 | 1.0 | 2.0 |
|---|---|---|---|
| ECE change | -7.3 | -2.3 | 5.8 |
| Default BA change | 0.0 | 0.0 | 0.0 |

Class separation, $d$

**(c) Ranking noise**

$s \rightarrow s + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, 1)$

Changes relative to shifted score s

| | 0.5 | 1.0 | 2.0 |
|---|---|---|---|
| AUC change | -3.9 | -6.9 | -8.0 |
| Headroom change | -2.5 | -4.9 | -7.6 |

Class separation, $d$

Change (pp): 15, 10, 5, 0, −5, −10, −15

Gray cells: loss level in pp at s. Blue/green: signed change, not benefit.

**ECE design ranges (pp; not CIs)**

$d = 0.5$: [-13.85, +7.29]

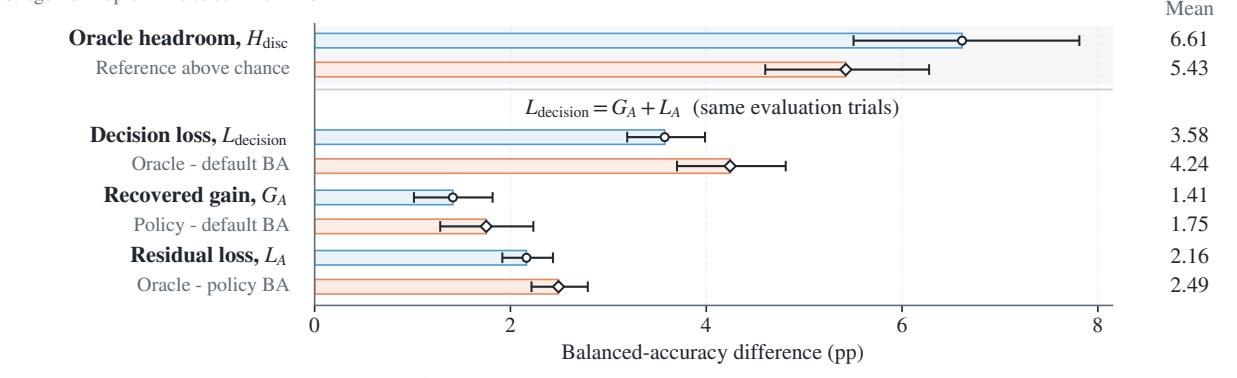
$d = 1$: [-10.32, +13.79]

ECE changes vary in sign across design cells.

$d = 2$: [-0.15, +15.56]

**(d) Protected EEG decision audit**

Stieger2021 | prefix-10 to suffix | n = 62




Bars and symbols: means. Whiskers: participant 95% CIs. Oracle uses labels.

Gain CIs use the post-confirmation bootstrap. Other CIs are descriptive. Oracle headroom is a separate reference.